Python ALA 2 = Removing Duplicates from List

1. What is a List?

A list in Python is a collection of items that are ordered and changeable.
It allows duplicate values.
Example:

my_list = [1, 2, 2, 3, 4, 4]

2. What are Duplicates?

Duplicate elements are values that appear more than once in a list.

Example:

In [1, 2, 2, 3, 4, 4]
2 and 4 are duplicates

3. Method 1: Using set()

```
my_list = [1, 2, 2, 3, 4, 4]

unique_list = list(set(my_list))

print(unique_list)
```

Explanation:

set() removes duplicate values
list() converts it back to list

Output:

[1, 2, 3, 4]

4. Method 2: Using Loop

```
my_list = [1, 2, 2, 3, 4, 4]
    unique_list = []

    for i in my_list:
        if i not in unique_list:
            unique_list.append(i)

    print(unique_list)
```

Explanation:

Loop checks each element
Adds only if not already present

Output:

[1, 2, 3, 4]

5. Method 3: Using Dictionary

```
my_list = [1, 2, 2, 3, 4, 4]

unique_list = list(dict.fromkeys(my_list))

print(unique_list)
```

Output:

[1, 2, 3, 4]

6. Conclusion

set() → fastest but unordered
Loop → easy to understand
dict.fromkeys() → keeps order